

All Pumped Up

New heat pump water heaters consume half the energy of conventional electric-resistance ones.

by John J. Tomlinson

About half of all residential water heaters in the United States use resistance heating elements inside a tank to convert electric energy into heat. The most efficient of these electric resistance water heaters (ERWH) is about 93% efficient, so it would seem that there is little room for further efficiency improvements to electric water heaters. But in fact, a heat pump can greatly increase a water heater's efficiency (see "Measuring Efficiency").

Today's residential heat pump water heater (HPWH) brings the efficiency of water heating to a level not possible with a conventional ERWH. Like an air conditioner or refrigerator, the heat pump water heater uses a compressor, an evaporator, a condenser, and an expansion device to transfer heat from one region to another. A refrigerator moves heat from inside the refrigerator to the house air. An air conditioner moves heat from inside the house to outside. In the case of a HPWH, heat is moved from the house, garage, or outside air into the water tank.

The advantage of the HPWH is based on the fact that it requires less work (less energy) to move heat from the surroundings into the water than it would to heat the water with a ERWH. This means that the Energy Factor (EF) of the heat pump water heater (defined as the ratio of the heat energy produced as hot water to the electrical energy needed to make this happen) can be greater than 1. A recent development is the drop-in heat pump water heater with a measured EF of 2.4 (a test model is shown in the photo above with Van Baxter, an ORNL researcher).



The key to this impressive performance is a small heat pump located on the top of the water tank. The pump consumes a relatively small amount of electricity to extract heat from the air surrounding the water heater (see Figure 1).

As the figure indicates, a conventional ERWH would consume 13.3 kWh of electrical energy to produce 64.3 gallons of hot water—the average daily hot water consumption for a typical U.S. household. This example assumes that 90% of the electrical energy is converted into the thermal energy (41,300 Btu) needed to heat water through a 77°F temperature change. A 90% efficiency represents the performance of an efficient ERWH. The average daily hot water consump-

tion (64.3 gallons) and the temperature change (77°F) are values used in the DOE test procedures for water heaters.

On the other hand, the drop-in HPWH shown on the right in Figure 1 requires only 4.9 kWh of electrical energy to accomplish the same task. Because a heat pump is part of the technology, most of the energy in the hot water—28,700 Btu in this case—is taken from the air surrounding the HPWH. This leaves only a small amount of electrical energy (4.9 kWh) that must be purchased. The heat removed from the surrounding air leaves it cool and dehumidified. (As with a room air conditioner, the moisture that is removed from the surrounding air is drained away as condensate, and a drain must be provided for the HPWH.) In the example above, the electrical energy consumption dropped from 13.3 kWh to 4.9 kWh to produce savings of 8.4 kWh each

day. These energy savings have been validated in the laboratory at the Oak Ridge National Laboratory (ORNL) Buildings Technology Center.

The heat comes from the surrounding air, even during the winter. If the HPWH is located in unconditioned space (for example, a garage or basement), there is little if any impact on the space conditioning system, depending on the configuration of the unconditioned space, and on its relation to the rest of the house. In a garage application, insulation in the common wall between the garage and the house and normal air infiltration into the garage tends to minimize the impact of the air source HPWH on the space-heating load. Unheated basements are good locations for the HPWH for the same reasons.

Table 1. Residential HPWHs

Manufacturer	Model Designation	Type	Heating Capacity (Btu per hour)*	Electrical	Refrigerant
ECR International	Water\$aver	Integral	4,500	208/230 Vac	R-134a
E-Tech	R106K4	Add-on	12,000	208/230 Vac	R-22
	WH-6BX-1	Add-on	6,000	208/230 Vac	R-22
Nyle Special Products	Nyletherm-110	Add-on	6,000 – 11,900**	120 Vac	R-401a
	Nyletherm-1	Add-on	8,000	208/230 Vac	R-401a
	Vaughan/Nyle Integrated-1	Integral	8,000	208/230 Vac	R-401a

*Nominal value over operation range of unit

**Sized and built to order

New Design, No Problems

Heat pump water heaters are not new (they were first commercialized in the mid-80s), but they are relatively unknown. Earlier HPWHs have not captured much of the residential water-heating market for a number of reasons. First, the high first cost leads to payback periods greater than five years. Second, the reputation for reliability has been poor. Customers expect a replacement water heater to last at least ten years—the minimum lifetime of a conventional water heater. Third, until now, HPWHs needed two trades for installation, a plumber and an HVAC technician. This complicated the installation procedure, increased the cost, and blurred the lines of responsibility for the installation. Fourth, in an effort to match the heating capacity of conventional water heaters, early designs used large components, which required evaporator flow rates of 400–600 CFM. This meant that the heat pump water heater needed to be located either in a ventilated space or in a large room. This constrained the adoption of the early HPWH as a replacement water heater.

The new drop-in residential HPWH resolves all these problems. Its market niche is the annual 3–4 million-unit replacement market for residential electric water heaters. It uses a 50-gallon hot water tank with an upper and lower heating element. Before the tank is insulated and covered, a condenser coil is wrapped around the lower portion of the tank; this coil provides most, if not all, of

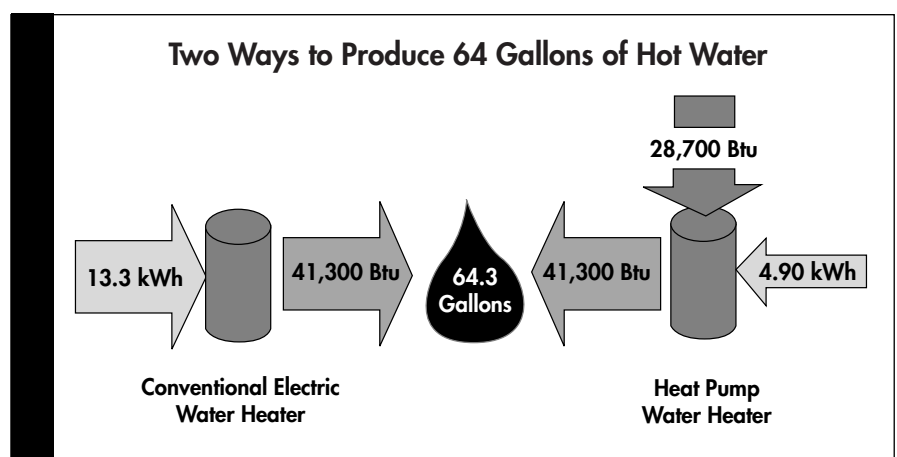


Figure 1. An HPWH can save 8.4 kWh each day for an average household.

the water heating. The drop-in design uses a small, robust refrigerator-like compressor with a capacity of about 3,500 Btu per hour, and a small, air source evaporator that is located next to the compressor on the top of the tank. A microprocessor controller with four sensors controls the source of water heating by selecting from the wraparound condenser, the upper heating element, or the lower heating element. The microprocessor controller and its logic determine the best mode of operation, depending on ambient, condensing, and evaporating temperatures; the setpoint for the water heater; and an envelope of acceptable operating conditions for the compressor. The drop-in heat pump water heater can heat water to 150°F under any set of ambient conditions of temperature and humidity seen by a conventional electric water heater.

Even low ambient temperatures with

high humidity, which might frost up the evaporative coil, work. A temperature sensor in the HPWH design senses evaporator coil temperature, and when frosting conditions are present, the HPWH compressor is turned off, and the lower resistance element in the tank is activated. Periodically, the compressor is turned on, and if evaporator temperatures do not fall to frosting conditions, the compressor remains on. Because it can revert to resistance heating in the case of extreme ambient temperatures, the HPWH can be applied to most situations where a conventional resistance water heater might otherwise be used.

The drop-in heat pump water heater is designed from the ground up to be a full replacement for a conventional electric resistance water heater. Because small components are used, the evaporator air flow requirement is low enough to permit the HPWH to work in a closet with

minimal ventilation, a basement, a garage, or most other locations in a house—wherever a conventional electric resistance water heater can be located. Locating the HPWH in a small, sealed closet, however, would not be desirable because the closet could become too cool, and the performance of the HPWH would suffer.

Accelerated-lifetime tests have shown the drop-in design to survive more than 7,000 compressor cycles—the equivalent of ten years of real-world use.

Ten States of Testing

DOE-sponsored field tests of the drop-in HPWH are being conducted with a number of utility partners across ten states. These utilities include the Tennessee Valley Authority, the Gulf Power Company, the Jackson Electric Membership Corporation, Northeast Utilities, Seattle City Light, the Wake Electric Cooperative, the East River Electric Power Cooperative, the Alabama Power Company, and the Lenoir City Utilities Board. Each utility identified one or more sites—homes having up to seven occupants—with electric resistance water heaters. In each test, a heavily instrumented drop-in

HPWH was installed in place of—and in the space vacated by—the resistance water heater. Some of the field test units were located in unconditioned spaces, such as a garage; some were located in semiconditioned spaces, such as a finished basement; and some were

average weekly electricity used at each site was calculated when the water heater was operating in electric resistance mode and in heat pump mode (that is, as designed). The results of this comparison are shown in Figure 2. The savings in electricity is shown as the difference between the upper set of data (resistance mode) and the lower set of data (heat pump mode). For any weekly hot water use, the HPWH uses about half of the electricity required by a comparable resistance water heater of the same size and in the same location.

It should be noted that per design, the heat pump water heater retains both heating elements as backup and this energy is included in the heat pump data in Figure 2. On average, the small compressor ran for 57 hours per week, the upper 4.5 kW element ran for 107 minutes per week, and the lower 4.5 kW element ran for 16 minutes per week.

Data from the field study were also used to measure the effect of the HPWH's location in the house on overall performance. Figure 3 shows that the coefficient of performance (COP), while quite high, drops as



(left) The Nyletherm-1, an example of an add-on water heater, is often wall-mounted. (right) The E-Tech R106 mounts to the top of a standard water heater.

located in conditioned spaces, such as a utility, laundry room, or closet.

The measured energy consumption of each heat pump water heater while operating as a resistance water heater was recorded as a function of the thermal energy delivered to the hot water that was used in the house each week. The HPWH operated at various times in HP and ER modes in order to compare performance. For the year, the

Measuring Efficiency

There are at least two efficiency metrics for heat pump water heaters: the Energy Factor (EF) and the coefficient of performance (COP). The EF is used to rate the efficiency of water heaters according to a uniform test procedure—the DOE 24-hour Simulated Use Test for water heaters. The test procedure prescribes the locations and required accuracies of measurement sensors, how often measurements must be made, initial water storage and inlet temperatures to the tank during water draws, and various additional conditions that must be maintained during the test. In the 24-hour Simulated Use Test, hot water is drawn from the tank at regular intervals and fixed amounts, and the

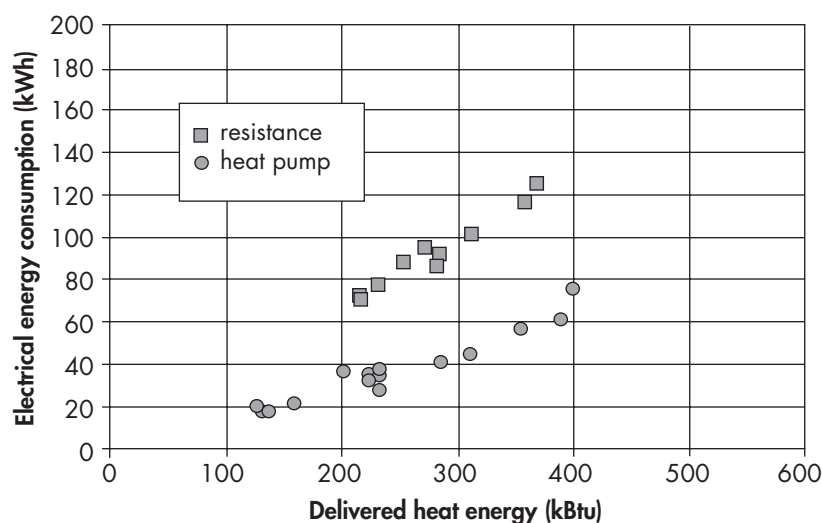
tank is allowed to recover over a 24-hour period. The EF is essentially the quotient of the heat energy produced by the water heater and the electrical (or gas energy) consumed. The EF can be used to compare all storage water heaters on an equal basis through a consistent laboratory test procedure.

Today's National Appliance Energy Conservation Act minimum-efficiency requirement for an electric storage water heater (50-gallon nominal capacity) is 0.86. Beginning January 20, 2004, the minimum efficiency for a 50-gallon model will rise to 0.90.

On the other hand, COP is a term that, for water heaters, can connote efficiency as may be experienced in the field. For COP to be a useful term, one

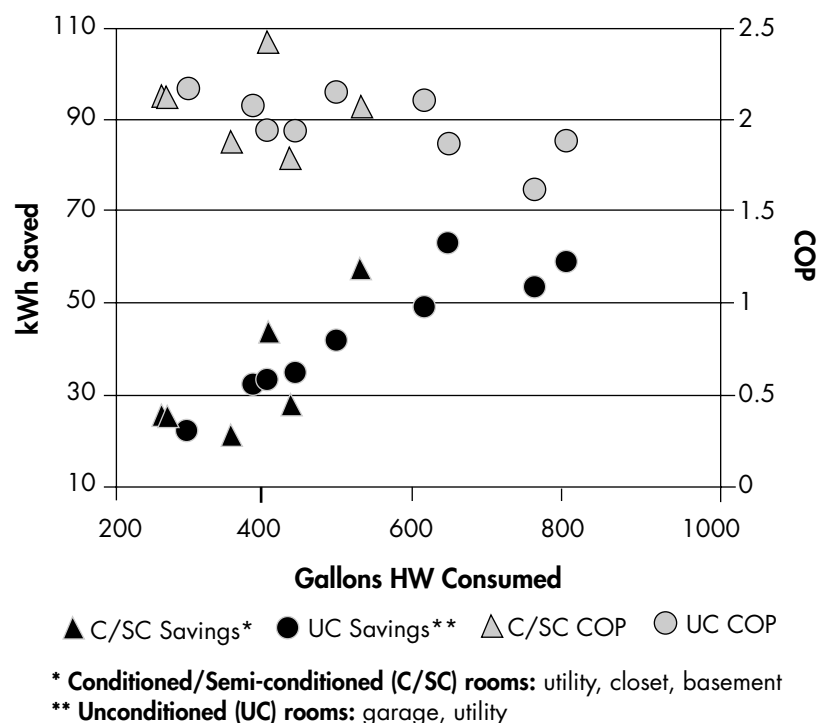
needs to be clear on the conditions under which a COP value is determined. In the national HPWH field study, we considered COP to be the total heat energy removed by hot water draws in one week divided by the total electrical energy used by the water heater over the same week (making allowances for the differences in the average tank temperature at the beginning and end of the week). Consequently, one would expect that the COP of a HPWH located in a garage in winter would be lower than the COP of the same HPWH in summer. The HPWH in winter experiences larger tank losses, a cooler entering water temperature, and typically cooler ambient temperatures—all factors that would contribute to a lower wintertime COP.

Resistance/Heat Pump Mode Performance Comparison



ORNL

Performance Affected by Weekly HW Use and by Siting



ORNL

Figure 2. (top) HPWHs consistently used half the electricity of electric resistance heaters in ORNL tests. **Figure 3.** (bottom) HPWH performance remains high wherever located.

weekly hot water use increases for all of the sites, irrespective of where the HPWH is located. However, energy savings (the real measure of value) increases with weekly hot water consumption. This indicates that, in

terms of energy saving, attaining a high COP through technology improvements is important, but siting the HPWH where hot water consumption is high is just as important. Moreover, the performance

of the HPWH, while affected by location, remains high whether the HPWH is located in a conditioned or in an unconditioned space.

How does the HPWH interact with the space conditioning system? There is no single answer to this question because the impact and benefits of the HPWH are application specific. In summer, basement dehumidification can be important, and a HPWH located in the basement could do the job. This would constitute a secondary benefit of a HPWH. In winter, if the space is unconditioned, the extra thermal load on the house heating system can be minor, depending on the level of insulation between the basement and the floor above.

Who Makes Residential Heat Pump Water Heaters?

Three manufacturers currently produce dedicated HPWHs (see Table 1).

Enviromaster (a subsidiary of ECR International) produces the design described in the field test and shown in the opener photo. This is an integral unit with a fixed condenser that is wrapped around the lower portion of the tank. The heating capacity of the compressor is about 4,000 Btu per hour, making it the smallest-capacity unit on the residential market.

Nyle Special Products produces the Nyletherm-1, an add-on HPWH. This unit is easily adapted to any electric resistance water heater. The heating output from the HPWH effectively replaces the operation of the heating element in the lower part of the tank. The Nyletherm-1 may be mounted on a wall and piped to the drain valve of the hot water storage tank. When the tank's lower thermostat calls for heat, the compressor, evaporator fan, and circulating pump are turned on. This draws cool water from the bottom of the tank, heats it using the condenser in the Nyletherm-1, and circulates it back to the storage tank. The heating capacity of the Nyletherm-1 is nominally 8,000 Btu per hour. Nyle Special Products also produces the Nyletherm-110. This unit is similar to the Nyletherm-1 but operates on 120 VAC (plug-in), making it quicker and easier to install. The

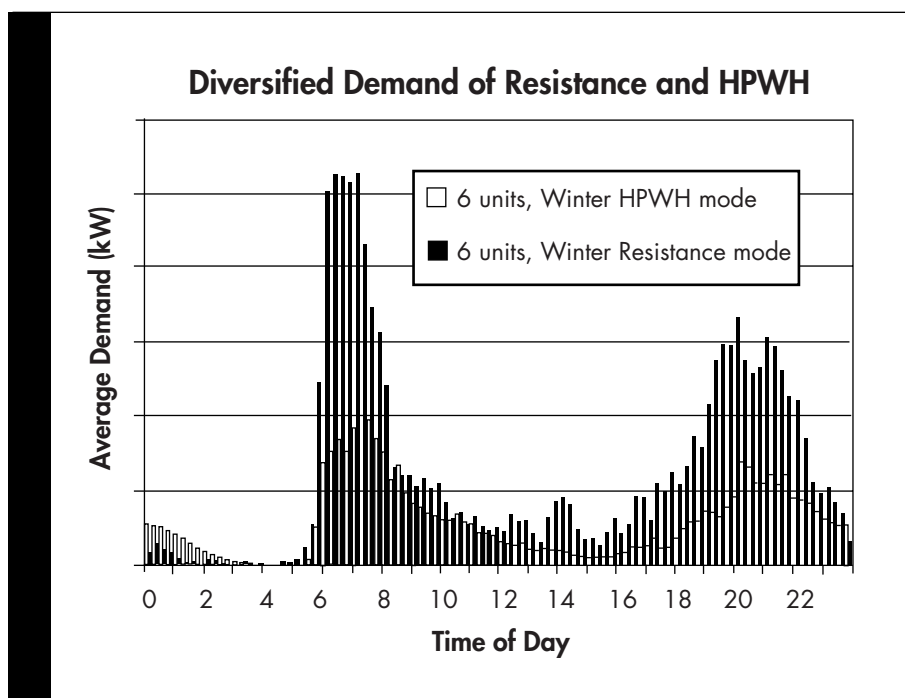


Figure 4. The magnitude of morning and evening peak demand drops significantly with HPWHs.

Nyletherm Integrated-1 is another configuration. Like all of the Nyle HPWHs, the Integrated-1 employs a small pump to circulate water between the tank and the condenser.

E-Tech (sold by Applied Energy Recovery Systems, Incorporated) manufactures two models for the residential market. Their Model R106K4 mounts to the top of a standard electric water heater and uses a small pump to circulate water from the tank inlet water line through the condenser on the HPWH and back to the storage tank at the hot water outlet. The heating capacity of the R106 is 12,000 Btu per hour. E-tech also produces a smaller HPWH, the Model WH-6BX-1. Like the Nyletherm-1, this model can be hung on a wall or located on the floor next to the water heater. With a heating capacity of 6,000 Btu per hour, this model is similar in capacity and size to the Nyletherm-1. Like the Nyletherm-1, the WH-6BX-1 is wired to operate using the thermostat of the lower thermostat but does not use the lower element itself. The manufacturer also provides an “energy savings” mode in which the upper element of the tank does not operate. The storage tank must be large enough to minimize hot water

runouts when the HPWH is connected in the energy savings mode.

Which One to Choose, Integral or Add-On?

Neither is “better” than the other. Each has advantages as well as disadvantages. The integral unit includes the tank, so unless the old tank has failed, it may not be water heater replacement time. On the other hand, full replacement of an existing resistance water heater with the drop-in design can easily be done by a plumber with no special skills. The add-on design is simple to install and would be a good choice particularly in cases where the existing tank is young and in good shape. As in the case of any water heater, a blanket provides added value.

Utility Impacts of the HPWH

The fact that all residential HPWHs rely on resistance heating for backup suggests that the HPWH with reduced overall energy consumption and high peak demand may represent an unfavorable load to electric utilities. The DOE field study and the ability to operate test units as conventional resistance water heaters and as heat pump water heaters

allowed ORNL to test this hypothesis. Test units at six of the test sites were operated alternately as resistance water heaters, and as heat pump water heaters for two five-week periods. Average electric demand data were collected and averaged to produce 15-minute diversified demand data (see Figure 4).

When the units were operated as resistance water heaters, two peak periods were noted as expected—one in the morning and one in the evening—coincident with hot water consumption patterns. The peak diversified morning demand was about 2.5 kW and the evening peak was about 1.5 kW. Although data from winter operation are shown, the same trend is present for water heating in the summertime.

When the units were returned to normal heat pump operation and the experiment continued, several changes were noted. First, the magnitude of the morning and evening peaks dropped significantly, and the maximum diversified demand fell to about 1 kW. Second, the demand late in the evening and in the early hours of the morning increased as the HPWH operated to heat the tank. This is a time when most utilities would have baseload capacity reserves, so that shifting some of the residential water heating load to this period is beneficial. Finally, most utilities see peak summer loads (which are driven principally by residential and commercial air conditioning) occurring in the afternoon when the electric demand for water heating is low. Despite this seeming mismatch in HPWH electric demand and utility peak demand, the HPWH benefit remains, although it becomes more subtle.

As shown in Figure 4, the afternoon demand for the HPWH is lower than the afternoon demand for the electric resistance water heater. Moreover, as the HPWH operates during the afternoon, it not only heats water but also provides summertime cooling and dehumidification. The cooling and dehumidification provided by the HPWH reduces the load on the home’s A/C and dehumidification system. In the case of the unit evaluated under the DOE field test program, the cooling/dehumidification benefit during HPWH operation is about 3,500 Btu per hour. For a central 10-SEER air conditioner to provide 3,500 Btu per

hour of cooling and dehumidification would require 350 watts of power. On the other hand, if the HPWH were used to offset the need for running an EnergyStar-rated dehumidifier in a space cooled by a 10-SEER air conditioner in the summer, the demand savings would be about 650 watts, depending on the size and efficiency of the dehumidifier. This reduction in electric load attributable to the HPWH would occur during the afternoon utility peak period, due to the HPWH's long recovery time. In summary, the HPWH benefits utilities by providing a significant reduction in on-peak electric demand.

Utility Interests

Some electric utilities promote and provide incentives for the installation of residential HPWHs. The largest HPWH utility program currently under way is Northeast Utilities' Hot-Shot program. Since it was introduced in 1997, the Northeast Utilities (NU) program has seen the installation of more than 4,700 add-on units in Connecticut and Maine. The program carefully screens applicants to make sure (1) that they have an electric water heater with a capacity of at least 50 gallons, (2) that the water heater is not located in conditioned space, so that the HPWH will not rob heat from the house and add to the space heating load in the winter, (3) that there is sufficient room to install the add-on HPWH, and (4) that the installation will not be in a garage, attic, or space that is not electrically heated. A successful applicant is assigned to an installation contractor who installs the HPWH. The cost to the homeowner, excluding any tank replacement, is \$150; NU picks up the rest of the costs. Overall, NU's experience with the HPWH has been positive. Not only do customers like the savings in operating cost; they also like the dehumidification that the HPWH provides. NU estimates that if the program were to terminate with the more than 4,700 HPWH that have already been installed, the electricity savings would be more than 200,000 MWh over the expected lifetime of the units.

HPWH Outlook

Today's residential HPWH designs are a significant improvement over earlier

designs. The tendency has been to move away from high-capacity units with short run times that chase the capacity of resistance elements to smaller-capacity units with longer run times. This design change has reduced the size requirements of heat exchangers and compressors, and these reductions have made possible the use of smaller evaporator fans, reduced the space needed for the heat pump components, and produced quieter systems overall. Today's designs are better suited to meet the needs and constraints of the residential consumer.

One challenge being faced by storage water tank manufacturers is that of standby losses, which largely determine the EF of water heaters. With a HPWH, the electrical energy needed to offset tank standby losses is about half of that needed for a resistance water heater. Therefore, efforts to improve tank insulation, which are essential for a high-EF resistance water heater, are a lower technological hurdle for HPWH designers. Today's designs resemble a refrigerator with a sealed system. The ECR HPWH design, for example, eliminates the need for a circulating pump (which also reduces callbacks). Designers of the Nyletherm model 110 have simplified the installation process by creating a 120VAC plug-in unit that can be mated to a wide variety of existing storage water heaters.

There's no denying that the first cost of a HPWH is two to three times greater than a resistance heater. But, depending on the application, the payback is rapid and reliable—in homes with several occupants and where electricity costs are at least in line with the U.S. average, payback is generally three to five years. The HPWH becomes even more attractive when one considers the cooling and dehumidification benefit. The DOE works with many partners to improve markets for both residential and commercial HPWHs. Efforts are under way to transfer field test results to industry, utilities, and consumers; to conduct both field and durability tests; and to organize customer groups for volume procurements.

The potential for significant market growth depends on increasing public awareness of the energy and cost savings provided by the HPWH and the ancillary benefits in terms of cooling and dehumidification. The DOE/EPA

Energy Star program does not include water heaters at the present time. Unlike every other technology, electric resistance water heating has hardly changed over the past 50 years. At best it can only store as heat all of the energy that enters it as purchased electricity. By using electricity and a heat pump cycle to move heat from the surroundings into the storage tank, the HPWH consumes 50%–60% less energy than the resistance water heater. At present, ERWHs comprise approximately half of all residential water heaters. If just 10% of these ERWHs were replaced by HPWHs, 5% of all electricity now devoted to electric water heating could be saved, with significant cost savings to consumers as well.



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